

Random Contrast Task Encoding Variants: Comprehensive Analysis Report

Date: December 29, 2024 **Author:** Can Konuk **Experiment Duration:** ~5 minutes

Executive Summary

This report documents experiments testing whether incorporating **randomly sampled hands** into the contrastive task encoding improves recognition model performance. Three new encoding strategies were tested:

1. **Random Augmented:** $= \text{mean}(\text{pos}) - \text{mean}(\text{neg}) + \lambda(\text{mean}(\text{combined}) - \text{mean}(\text{random}))$
2. **Positive vs Random:** $= \text{mean}(\text{pos}) - \text{mean}(\text{random})$
3. **Triple Contrast:** $\text{concat}([\text{pos-neg}, \text{pos-random}, \text{neg-random}])$

Key Finding: The random contrast variants **do NOT improve** over the standard contrastive encoding. The baseline enhanced card encoder with standard task encoding achieves the best performance ($R@5=0.657$). Random hand comparisons slightly **degrade** performance, suggesting that task-specific positive-negative contrast is more informative than general distinctiveness from random hands.

1. Experimental Rationale

Motivation

The standard contrastive encoding ($= \text{mean}(\text{pos}) - \text{mean}(\text{neg})$) captures what distinguishes positive examples from negative examples for a specific rule. We hypothesized that adding comparisons with randomly sampled hands might: - Help distinguish “what makes these hands special” vs just “what makes positives different from negatives” - Provide a baseline for comparison that isn’t rule-specific - Capture features that make both positive and negative examples stand out from typical random hands

Variants Tested

Variant	Formula	Hypothesis
Random Augmented	$= (\text{pos-neg}) + \lambda(\text{combined-random})$	Combines task-specific contrast with general distinctiveness
Positive vs Random	$= \text{pos} - \text{random}$	Focus on what makes positive hands special overall
Triple Contrast	$\text{concat}(\text{pos-neg}, \text{pos-random}, \text{neg-random})$	Rich representation with all three contrasts

2. Results

Main Results Table (Sorted by $R@5$)

Variant	$R@5$	$R@10$	MRR	Conv. Epoch
baseline__enhanced	0.657 ± 0.02	0.788	0.466 ± 0.01	55
triple_contrast_standard	0.644 ± 0.03	0.792	0.454 ± 0.01	54
random_augmented_lambda0.5	0.640 ± 0.05	0.786	0.461 ± 0.01	52
baseline_standard	0.635 ± 0.02	0.793	0.459 ± 0.01	48
random_augmented_lambda1.0	0.627 ± 0.03	0.791	0.455 ± 0.01	50

Variant	R@5	R@10	MRR	Conv. Epoch
pos_vs_random_standard	0.620±0.02	0.804	0.462±0.01	53
pos_vs_random_enhanced	0.619±0.04	0.789	0.457±0.01	56
enhanced+random_augmented	0.619±0.02	0.791	0.454±0.01	53
random_augmented_50random	0.606±0.04	0.783	0.448±0.01	55
enhanced+attention+triple	0.604±0.02	0.793	0.459±0.01	57
triple_contrast_50random	0.600±0.07	0.797	0.459±0.02	60
triple_contrast_enhanced	0.588±0.03	0.776	0.456±0.02	51

Task Encoder Comparison

Encoder Type	Mean R@5	Mean MRR	Δ vs Standard
Standard	0.646	0.462	—
Random Augmented	0.623	0.455	-3.6%
Positive vs Random	0.620	0.460	-4.0%
Triple Contrast	0.609	0.457	-5.7%

3. Analysis

3.1 Why Random Contrast Doesn't Help

Key Insight: The task-specific contrast (pos vs neg) already captures the essential discriminative information. Random hands add noise rather than signal because:

1. **Task-Specificity Matters Most:** The rule itself defines what makes a hand positive or negative. The contrast $\text{pos} - \text{neg}$ directly encodes this rule-specific distinction.
2. **Random Hands Are Uninformative:** Random hands don't follow any rule, so comparing with them doesn't provide useful information about the target rule.
3. **Noise Injection:** Random hands introduce variance in the embedding space that doesn't correlate with the primitives needed to solve the rule.
4. **Diluted Signal:** In Triple Contrast, the useful (pos-neg) signal gets diluted by the less informative (pos-random) and (neg-random) components.

3.2 Lambda Parameter Analysis

For Random Augmented encoding with different λ values: $\lambda = 0.5$: R@5=0.640 (5% below baseline) - $\lambda = 1.0$: R@5=0.627 (7% below baseline)

Higher λ (more weight on random contrast) **degrades** performance, confirming that the random contrast term adds noise.

3.3 Number of Random Hands

Testing with different numbers of random hands (25 vs 50):
- 25 random hands: Generally performs better
- 50 random hands: Slightly worse (more noise, less focused embedding)

More random hands doesn't help; it likely increases the variance in the random mean embedding.

3.4 Interaction with Card Encoders

- **Standard card encoder** works better with random contrast variants
 - **Enhanced card encoder** shows degradation with random contrast
 - This suggests the enhanced features may conflict with the noise from random comparisons
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4. Comparison with Architectural Variants

Combining with findings from the architectural variants comparison:

Architecture	Best R@5	Notes
Enhanced card + Standard task	0.657	Best overall
Standard card + Standard task	0.635	Baseline
Standard card + Triple	0.644	Random contrast doesn't hurt much here
Enhanced + Random variants	0.619	Combination hurts performance

Recommended Configuration

```
model = RecognitionModelVariant(  
    grammar=grammar,  
    card_encoder_type='enhanced',      # Adds color + rank value  
    hand_encoder_type='mean',         # Simple pooling works best  
    task_encoder_type='standard',     # = pos - neg (NOT random variants)  
    prediction_head_type='sigmoid',   # Much better than embedding heads  
    loss_type='bce'                   # Better than focal loss  
)
```

5. Theoretical Implications

For Program Induction

The failure of random contrast encodings suggests that: 1. **Task-specific supervision is key:** The recognition network needs to learn what distinguishes solutions from non-solutions, not what distinguishes “interesting” hands from random ones. 2. **Contrastive learning works when contrasts are meaningful:** Pos vs neg hands have meaningful contrast (the rule); pos/neg vs random doesn't.

For Self-Explanation

This relates to the self-explanation literature: - **Active processing of correct-incorrect contrasts** (what chi-explanation captures) is more valuable than passive comparison with baseline - The “rules” people induce are about discriminating categories, not about general “interestingness”

6. Bug Fixed During Experiments

Critical Bug: The `compute_loss` method was looking for `task.rule.program` which doesn't exist. - Catalogue rules have `primitives_used` as a list - Pretraining rules have `expected_program` as a string

Fix: Updated `compute_loss` to try multiple sources: 1. `task.primitives_used` (direct attribute) 2. `task.rule.primitives_used` (catalogue rules) 3. `task.rule.expected_program` (pretraining rules, parsed) 4. `task.rule.program` (legacy fallback)

7. Conclusions

1. **Random contrast encodings do not improve recognition model performance**
2. **Standard contrastive encoding (= pos - neg) is optimal**
3. **Task-specific contrast is more informative than comparison with random hands**
4. **Enhanced card encoder remains the best architectural choice**
5. **Simpler is better:** Adding complexity (more contrasts, more random hands) hurts performance

Key Takeaway

For recognition networks in program induction, focus on **task-specific discriminative features** rather than general distinctiveness. The pos-neg contrast captures exactly what the model needs to learn: which primitives distinguish positive examples from negative ones for each rule.

8. Files Created/Modified

File	Purpose
<code>dreamcoder_core/recognition_variants.py</code>	Added RandomAugmented, PositiveVsRandom, TripleContrast encoders; Fixed <code>compute_loss</code> bug
<code>experiments/run_random_contrast_variants.py</code>	Experiment runner for random contrast variants
<code>results_random_contrast_*/</code>	Experiment results and reports

Appendix: Experimental Setup

- **Training Rules:** 44 pre-training rules (5-fold cross-validation)
- **Test Rules:** 45 catalogue rules
- **Hand Size:** 5-6 cards per example
- **Examples per Task:** 50 (25 positive, 25 negative)
- **Random Hands per Task:** 25 or 50
- **Training:** Adam optimizer, lr=0.001, 100 epochs, patience=15
- **Architecture:** 59 primitives, 64-dim embeddings, sigmoid prediction head